

R Laboratory - Session 3

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Additional comments explaining the code in a more detailed way are not reported here, but are available on the corresponding .R source code

We will analyze a simulated binary-response dataset (`logistic_data.csv`) using Bayesian Bernoulli regression. The data consist of observations $D = \{(y_i, x_i)\}_{i=1}^N$ where $y_i \in \{0, 1\}$ is a binary outcome and x_i is a real-valued explanatory variable. The model is defined as :

$$y_i | \beta_0, \beta_1 \sim \text{Bernoulli}(\pi_i)$$

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_i$$

which is equivalent to

$$\pi_i = \frac{\exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)}$$

The independent priors are given by :

$$\beta_0 \sim \mathcal{N}(0, 3^2), \quad \beta_1 \sim \mathcal{N}(0, 3^2)$$

(1) Posterior formulation and implementation

To write the joint posterior distribution $p(\beta_0, \beta_1 | D)$, we can apply the Bayes theorem :

$$p(\beta_0, \beta_1 | D) \propto p(D | \beta_0, \beta_1) p(\beta_0) p(\beta_1)$$

First, we write the likelihood of the data. We have that the observations y_i are independent conditional on the parameters, so the joint likelihood is the product of individual Bernoulli probabilities :

$$p(D | \beta_0, \beta_1) = \prod_{i=1}^N \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$$

By substituting the logistic link function $\pi_i = \frac{\exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)}$ and $1 - \pi_i = \frac{1}{1 + \exp(\beta_0 + \beta_1 x_i)}$, the likelihood simplifies to :

$$p(D | \beta_0, \beta_1) = \prod_{i=1}^N \frac{(\exp(\beta_0 + \beta_1 x_i))^{y_i}}{1 + \exp(\beta_0 + \beta_1 x_i)} = \prod_{i=1}^N \frac{\exp(y_i(\beta_0 + \beta_1 x_i))}{1 + \exp(\beta_0 + \beta_1 x_i)}$$

Then, the independent Gaussian priors for β_0 and β_1 with mean 0 and variance $\sigma^2 = 9$ are given by :

$$p(\beta_0) p(\beta_1) = \left(\frac{1}{\sqrt{2\pi \cdot 9}} \exp\left(-\frac{\beta_0^2}{18}\right) \right) \left(\frac{1}{\sqrt{2\pi \cdot 9}} \exp\left(-\frac{\beta_1^2}{18}\right) \right) \propto \exp\left(-\frac{\beta_0^2 + \beta_1^2}{18}\right)$$

We get the posterior formulation up to a proportionality constant by multiplying the likelihood and the priors :

$$\Rightarrow p(\beta_0, \beta_1 | D) \propto \left[\prod_{i=1}^N \frac{\exp(y_i(\beta_0 + \beta_1 x_i))}{1 + \exp(\beta_0 + \beta_1 x_i)} \right] \exp\left(-\frac{\beta_0^2 + \beta_1^2}{18}\right)$$

This posterior distribution does not have a standard closed analytical form, which justifies the use of MCMC methods to sample from it. To implement this model, we chose to use JAGS with the `rjags` package in R.

JAGS parameterizes the normal distribution using precision ($\tau = 1/\sigma^2$) instead of variance (σ^2). Therefore, the priors $\mathcal{N}(0, 3^2)$ are strictly implemented as `dnorm(0, 1/9)` in the model definition.

The data preparation and the JAGS model code are defined like this :

```

1 # dataset
2 logistic_data <- read.csv("logistic_data.csv")
3 y <- logistic_data$y
4 x <- logistic_data$x
5 N <- length(y)
6
7 # defining the JAGS model as a string
8 model_string <- "
9 model {
10   for (i in 1:N) {
11
12     # Likelihood
13     y[i] ~ dbern(pi[i])
14     logit(pi[i]) <- beta0 + beta1 * x[i]
15   }
16
17   # priors
18   beta0 ~ dnorm(0, 1/9)
19   beta1 ~ dnorm(0, 1/9)}
20 "
21
22 # data list for JAGS
23 jags_data <- list(
24   y = y,
25   x = x,
26   N = N)

```

(2) MCMC fitting and diagnostics

To estimate the posterior distribution, we ran the JAGS model using 4 separate MCMC chains. To be sure to have a robust exploration of the parameter space and avoid local traps, the chains were initialized with dispersed starting values located in the four opposite quadrants of the (β_0, β_1) plane: $(-5, -5)$, $(5, 5)$, $(-5, 5)$, and $(5, -5)$.

The configuration, burn-in, and sampling were executed using the following R code:

```

1 # 1
2 # We define different and spaced initial values for the 4 chains
3 # and we place them in the four quadrants to be sure that the entire parameter
   space is explored
4 inits_list <- list(
5   list(beta0 = -5, beta1 = -5),
6   list(beta0 = 5,  beta1 = 5),
7   list(beta0 = -5, beta1 = 5),
8   list(beta0 = 5,  beta1 = -5))
9
10 # 2
11 # compiling the JAGS model
12 # n.adapt = 1000 is used by JAGS to tune the samplers before the actual burn-in
13 jags_model <- jags.model(

```

```

14 file = textConnection(model_string),
15 data = jags_data,
16 inits = inits_list,
17 n.chains = 4,
18 n.adapt = 1000)
19
20 # 3
21 # burn-in period
22 # we remove the first 2000 iterations to be sure that the chains have no
    influence from their starting points
23 burnin <- 2000
24 update(jags_model, n.iter = burnin)
25
26 # 4
27 # sampling period
28 # we draw 5000 samples / chain, meaning our final posterior pool will have
    20000 valid samples
29 n_samples <- 5000
30 posterior_samples <- coda.samples(
31   model = jags_model,
32   variable.names = c("beta0", "beta1"),
33   n.iter = n_samples)
34
35 # 5
36 # diagnostics
37
38 # Gelman-Rubin diagnostic (R-hat)
39 # the values close to 1 (strictly < 1.05) indicate that all chains converged to
    the same distribution
40 rhat <- gelman.diag(posterior_samples)
41
42 # Effective Sample Size
43 ess <- effectiveSize(posterior_samples)

```

Each chain was run for a warm-up (burn-in) period of 2000 iterations to guarantee that they completely forgot their extreme initial states. After the burn-in, we kept 5000 sampling iterations per chain, giving us a combined posterior pool of 20 000 valid samples.

Based on the numerical diagnostics and visual inspections, the convergence and mixing of the chains are very good :

Gelman-Rubin Diagnostic (R-hat):
Potential scale reduction factors:

	Point est.	Upper C.I.
beta0	1	1
beta1	1	1

Multivariate psrf
1

Effective Sample Sizes (ESS):

beta0	beta1
11438.01	11529.43

- The visual inspection of the trace plots (Figure 1) shows an immediate convergence. All four chains overlap perfectly in a dense, structureless band without wandering trends, indicating that the Markov chains are exploring the high-probability region efficiently and symmetrically.

- The potential scale reduction factor ($\hat{R}$) evaluates the ratio of between-chain variance to within-chain variance. For both β_0 and β_1 , the computed $\hat{R}$ point estimate is exactly 1.00 (with an upper confidence interval of 1.00). Furthermore, the multivariate PSRF is 1.00. Since $\hat{R} < 1.05$ is the standard threshold, this proves that the chains have reached their stationary distribution.
- The autocorrelation function (ACF) plots (Figure 2) show an very rapid decay, dropping to zero essentially after the first lag. Because the successive samples are already highly independent, applying a thinning factor is completely unnecessary and would only discard valuable information. Therefore, no thinning was applied.
- Given the low autocorrelation, the Effective Sample Sizes computed by the `coda` package are remarkably high: ~ 11438 for β_0 and ~ 11529 for β_1 . Representing more than half of the total 20 000 samples, these ESS values confirm that our posterior summaries will be highly stable and reliable.

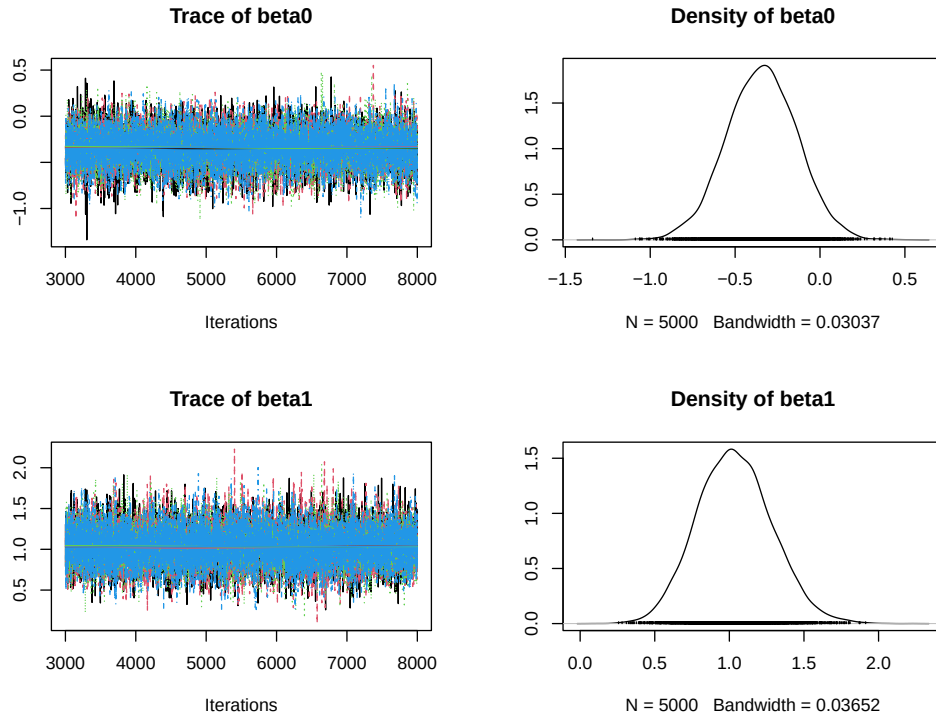


Figure 1: Trace plots and marginal posterior densities for β_0 and β_1 . The four independent chains perfectly overlap, showing optimal mixing properties.

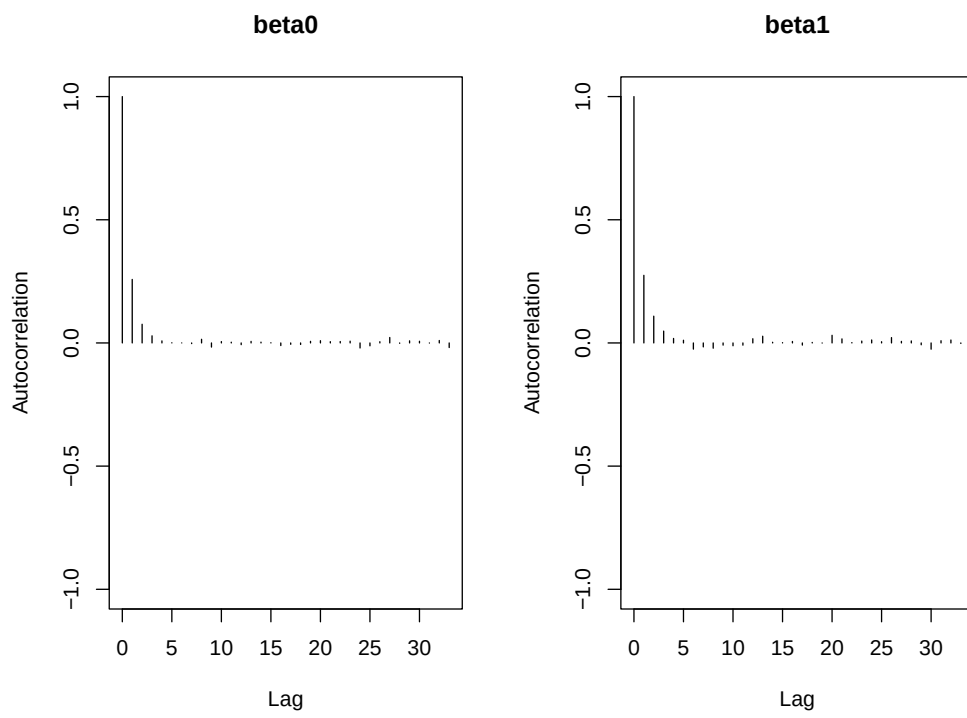


Figure 2: Autocorrelation function (ACF) for β_0 and β_1 . The rapid drop to near-zero at lag 1 justifies the absence of a thinning factor.

(3) Posterior inference

Using the 20 000 samples generated by our MCMC algorithm, we report the numerical posterior summaries for the intercept β_0 and the covariate effect β_1 .

```
1 # 1
2 # combining the 4 chains into a single matrix of 20 000 samples
3 samples_matrix <- do.call(rbind, posterior_samples)
4 b0 <- samples_matrix[, "beta0"]
5 b1 <- samples_matrix[, "beta1"]

1 # 4
2 # computing the posterior correlation
3 correlation <- cor(b0, b1)
4 cat("\nPosterior correlation between beta0 and beta1 : ", correlation, "\n")
```

The numerical extraction gives us the following terminal output :

```
beta0 (intercept):
Mean      = -0.3394789
Median    = -0.3382395
SD        = 0.2089454
95% CrI   = [ -0.7513545 , 0.06679428 ]
```

```
beta1 (covariate effect):
Mean      = 1.033926
Median    = 1.025284
SD        = 0.2500256
95% CrI   = [ 0.5663163 , 1.546488 ]
```

```
Posterior correlation between beta0 and beta1 : -0.2149283
```

The rounded summaries are grouped in the table 1 :

Parameter	Mean	Median	Standard Deviation	95% Credible Interval
β_0 (Intercept)	-0.34	-0.34	0.21	[-0.75, 0.07]
β_1 (Covariate effect)	1.03	1.03	0.25	[0.57, 1.55]

Table 1: Posterior summaries for the Bayesian logistic regression parameters.

Based on these numerical results and the corresponding visualizations (Figure 3), we can draw several conclusions about the covariate effect and the parameter dependencies :

- The estimated effect of the explanatory variable x is strictly positive, with a posterior mean of 1.03. This indicates that an increase in x systematically increases the log-odds (and so the probability) of the outcome $y = 1$.
- The standard deviation for β_1 is relatively narrow ($SD \approx 0.25$). More importantly, the 95% credible interval $[0.57, 1.55]$ does not contain zero, nor does it even approach it. This provides evidence that the covariate x has a statistically significant and meaningful impact on the response.
- We observe a moderate negative posterior correlation between the two parameters ($r \approx -0.21$). Visually, this is confirmed by the slight downward tilt of the elliptical cloud in the joint posterior scatter plot (Figure 3, right panel). This behavior is standard in linear and logistic models : a steeper estimated slope (β_1) tends to be mathematically compensated by a lower intercept (β_0) to maintain a good fit to the observed data mass.

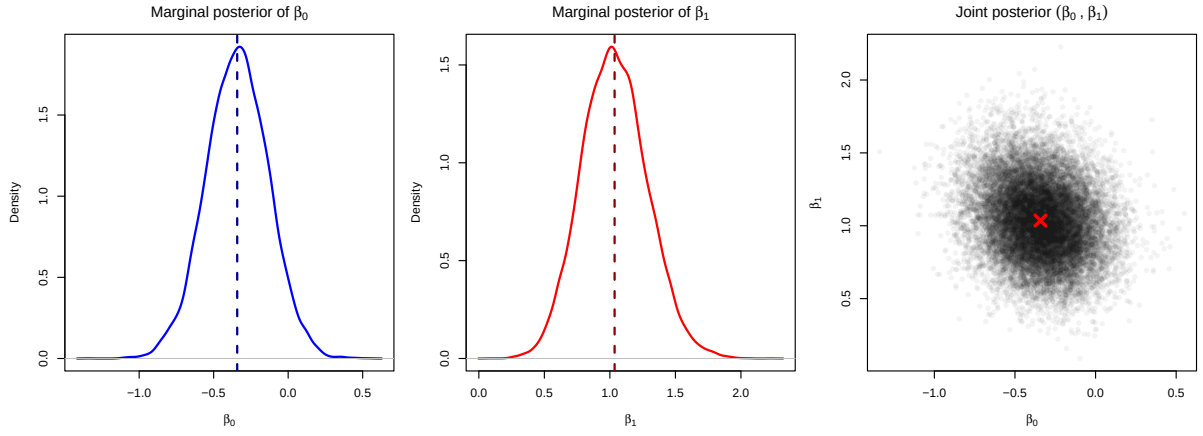


Figure 3: Left and Center : Marginal posterior density distributions for β_0 and β_1 with their respective means marked by dashed lines. Right : Joint posterior samples scatter plot in the (β_0, β_1) plane, showing the slight negative correlation.

(4) Posterior predictive inference

To evaluate the predictive behaviour of our model, we choose a new covariate value $x_{new} = 1.0$. This value falls comfortably within the observed data range and represents a positive response scenario based on the covariate effect established in the previous section.

The following R code was executed to derive the predicted probability parameter π_{new} and the simulated predictive observation y_{new} :

```
1 # 1
2 # choosing a new covariate value
3 x_new <- 1.0
4
5 # 2
6 # computing the posterior draws for pi_new
7 # using the logistic function: pi = exp(beta0 + beta1 * x) / (1 + exp(beta0 +
  beta1 * x))
8 eta_new <- b0 + b1 * x_new
9 pi_new <- exp(eta_new) / (1 + exp(eta_new))
10
11 # 3
12 # simulating the posterior predictive draws for y_new
13 # for each sampled probability pi_new, we draw exactly one Bernoulli trial (of
  size = 1)
14
15 set.seed(2107530) # for reproducibility
16 y_new_sim <- rbinom(n = length(pi_new), size = 1, prob = pi_new)
17
18 # posterior predictive probability P(y_new = 1 | x_new, D)
19 prob_y_new_1 <- mean(y_new_sim)
20 # law of total expectation => this is equivalent to mean(pi_new), but we
  simulate it explicitly so it proves the generative process
```

The terminal evaluates the corresponding summaries :

Posterior summaries for $\pi_{new} = P(y_{new} = 1 \mid x_{new}, \beta_0, \beta_1)$:

```
Mean      = 0.663925
Median    = 0.6659415
SD        = 0.06321205
95% CrI   = [ 0.5367918 , 0.7815902 ]
```

Posterior predictive probability $P(y_{new} = 1 \mid x_{new}, D)$: 0.6626

The posterior summaries for this probability parameter π_{new} are :

- Posterior mean : $\mathbb{E}[\pi_{new} \mid D] \approx 0.664$
- 95% Credible Interval : [0.537, 0.782]

To obtain the actual posterior predictive probability $P(y_{new} = 1 \mid x_{new}, D)$, we incorporate the sampling uncertainty by simulating a new Bernoulli observation $y_{new}^{(t)} \sim \text{Bernoulli}(\pi_{new}^{(t)})$ for each sampled probability. The empirical proportion of positive outcomes in our simulated predictions gives :

$$P(y_{new} = 1 \mid x_{new}, D) \approx 0.663$$

As dictated by the law of total expectation, this simulated predictive probability perfectly aligns with the theoretical posterior mean of the parameter π_{new} .

Figure 4 shows the posterior distribution of π_{new} . The density is smoothly distributed around ~ 0.66 , and the bounds of the 95% credible interval prove that for $x = 1.0$, the probability of a positive outcome $y = 1$ is strictly and confidently greater than 50%.

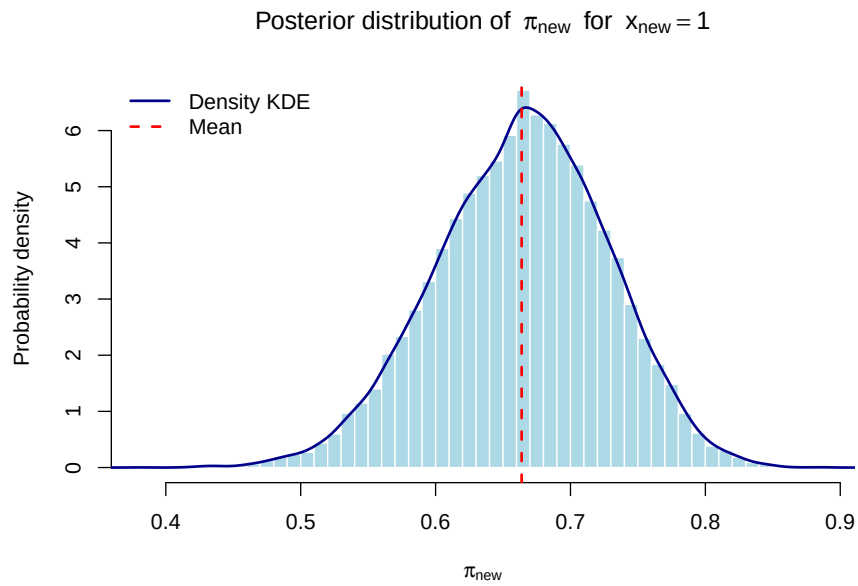


Figure 4: Histogram and Kernel Density Estimation (KDE) of the posterior distribution for π_{new} given $x_{\text{new}} = 1.0$. The red dashed line marks the posterior mean.

(5) Model comparison

To formally determine whether the covariate x_i improves the predictive performance of our system, we compare the full logistic model (M_1) against a simpler intercept-only baseline model (M_0).

The most appropriate Bayesian criterion for comparing models fitted via MCMC in JAGS is the Deviance Information Criterion (DIC). The DIC evaluates the trade-off between model fit (mean deviance) and model complexity (effective number of parameters, or penalty p_D). Models with smaller DIC values are preferred.

Using the `dic.samples` function natively provided by the `rjags` package, we computed the DIC for both models over 5000 iterations :

```
1 # 1
2 # definition of the M0 model (intercept-only) as a string
3 # --> we drop beta1 totally
4
5 model_string_m0 <- "
6 model {
7   for (i in 1:N) {
8     y[i] ~ dbern(pi[i])
9     logit(pi[i]) <- beta0
10  }
11
12  # Prior (using precision tau = 1/9)
13  beta0 ~ dnorm(0, 1/9)}
14 "
15
16 # 2
17 # initializing and compiling M0
18 inits_list_m0 <- list(
19   list(beta0 = -5),
20   list(beta0 = 5),
21   list(beta0 = -2),
22   list(beta0 = 2))
23
24 jags_model_m0 <- jags.model(
25   file = textConnection(model_string_m0),
26   data = jags_data,
27   inits = inits_list_m0,
28   n.chains = 4,
29   n.adapt = 1000)
30
31 # M0 burn-in
32 update(jags_model_m0, n.iter = 2000)
33
34 # 3
35 # computing DIC for the 2 models
36 # we use dic.samples() which is native in the rjags library
37 n_dic_samples <- 5000
38
39 dic_m0 <- dic.samples(jags_model_m0, n.iter = n_dic_samples)
40
41 # 'jags_model' is the original M1 model we compiled in the question 2
42 dic_m1 <- dic.samples(jags_model, n.iter = n_dic_samples)
43
44 # difference (M1 - M0)
45 diff_dic <- diffdic(dic_m1, dic_m0)
```

Model M0 (intercept-only) :
Mean deviance : 166.1
penalty 0.9626

Penalized deviance : 167.1

Model M1 (Full model) :

Mean deviance : 145.4

penalty 1.991

Penalized deviance: 147.4

Difference in DIC :

Difference : -19.65049

Sample standard error : 8.244768

The results are summarized in the table 2 :

Model	Mean Deviance	Penalty (p_D)	Penalized Deviance (DIC)
M_0 (Intercept-only)	166.1	0.96	167.1
M_1 (Full model)	145.4	1.99	147.4

Table 2: Deviance Information Criterion (DIC) comparison between the intercept-only model (M_0) and the full model including the covariate x_i (M_1).

- The penalty term accurately reflects the theoretical complexity of our models. Model M_0 has an estimated effective number of parameters of 0.96 (representing only β_0), while M_1 has a penalty of roughly 1.99 (representing both β_0 and β_1).
- The model M_1 has a significantly lower Mean Deviance (145.4 vs 166.1), meaning it fits the observed data much better than M_0 .
- Evaluating the final DIC, the full model M_1 scores 147.4, whereas the baseline M_0 scores 167.1. The difference is strictly negative ($\Delta DIC \approx -19.65$). In standard Bayesian practice, a DIC reduction greater than 10 rules out the worse model decisively. Therefore, M_1 is massively supported by the data over M_0 . We conclude that the inclusion of the covariate x_i improves the predictive performance of the system and is strictly necessary to explain the variance of the binary response.

(6) Discussion

Based on the Bayesian logistic regression analysis performed in the previous sections, we can summarize our findings by addressing the important aspects of the model's performance and statistical validity :

- The MCMC chains undoubtedly converged to the target posterior distribution. The trace plots showed immediate and perfect overlap, and the autocorrelation dropped to zero at lag 1. This is quantitatively confirmed by a Gelman-Rubin diagnostic ($\hat{R}$) of exactly 1.00 and very high Effective Sample Sizes (ESS > 11400 out of 20,000 samples).
- The posterior uncertainty is relatively narrow and completely manageable. With a sample size of $N = 120$, the model was able to constrain the standard deviations of both the intercept (≈ 0.21) and the slope (≈ 0.25) to low values, resulting in tight and highly informative 95% credible intervals.

- The effect of the explanatory variable x is credibly and strictly different from zero. The posterior mean for β_1 is ≈ 1.03 , and its 95% credible interval is $[0.57, 1.55]$. Because zero is far outside this interval, we have conclusive evidence that x is a significant driver of the probability of observing a positive outcome ($y = 1$).
- The posterior predictive probabilities are fully compatible with the observed data. As demonstrated for $x_{new} = 1.0$, the model rigorously propagated the parameter uncertainty into a smooth predictive probability distribution ($\mathbb{E}[\pi_{new}|D] \approx 0.66$). This correctly mirrors the underlying generative Bernoulli process and avoids overconfident predictions.
- The full model (M_1) is heavily preferred over the intercept-only baseline model (M_0). The Deviance Information Criterion (DIC) for M_1 (147.4) is significantly lower than for M_0 (167.1). With a massive DIC difference of roughly -19.65 , standard Bayesian model selection criteria overwhelmingly support the inclusion of the covariate x_i to explain the data.